TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

AI.310

Robot Localization and Mapping

COURSE OVERVIEW

Knowing where you are and what is around you: localization, mapping, and SLAM.

ROBOT LOCALIZATION AND MAPPING

1

SESSION

 60 MINUTES

ORIENTATION: WHERE AM I?

KEY TOPICS

- Localization vs. mapping
- The localization problem
- Map representations
- SLAM overview

TEACHING FOCUS

Ground the robot in its environment.

▲ **KEY TAKEAWAY** Navigation starts with knowing where you are.

ROBOT LOCALIZATION AND MAPPING

2

SESSION

60 MINUTES

LOCALIZATION

KEY TOPICS

- Monte Carlo localization
- Particle filters
- EKF localization
- Global vs. local localization

TEACHING FOCUS

Estimate robot pose from sensors.

▲ **KEY TAKEAWAY** Filters track pose over time.

ROBOT LOCALIZATION AND MAPPING

3

SESSION

60 MINUTES

MAPPING

KEY TOPICS

- Occupancy grid maps
- Feature maps
- Topological maps
- Map building

TEACHING FOCUS

Represent the world for a robot.

KEY TAKEAWAY Maps are the robot's memory of space.

ROBOT LOCALIZATION AND MAPPING

4

SESSION

60 MINUTES

SLAM

KEY TOPICS

- Simultaneous localization and mapping
- EKF SLAM
- FastSLAM
- Graph-based SLAM

TEACHING FOCUS

Build a map while locating in it.

▲ KEY TAKEAWAY SLAM solves the chicken-and-egg problem.

5

SESSION

60 MINUTES

LOOP CLOSURE & RELOCALIZATION

KEY TOPICS

- Detecting revisited places
- Loop closure
- Kidnapped robot problem
- Global localization

TEACHING FOCUS

Correct drift and recover from loss.

KEY TAKEAWAY Loop closure keeps maps consistent.

ROBOT LOCALIZATION AND MAPPING

6

SESSION

60 MINUTES

LOCALIZATION & MAPPING & REVIEW

KEY TOPICS

- SLAM demo
- Course recap and assessment
- Toward robotics planning (AI.311)
- Mapping exercise

TEACHING FOCUS

Run SLAM on a mobile robot.

KEY TAKEAWAY A mapped world is a navigable world.